



Anti-ulcerogenic effect of Zuojin Pill against ethanol-induced acute gastric lesion in animal models



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ABSTRACT

Ethnopharmacological relevance: Zuojin Pill (ZJP), a traditional Chinese medicinal decoction, contains two herbal drugs: *Coptis chinensis* Franch. and *Tetradium ruticarpum* (A. Juss.) Hartley in the ratio of 6:1 (w/w). In this study, ZJP was evaluated for its gastroprotective potential against mucosal lesions induced by ethanol in mice.

Materials and methods: 50 mice were assigned to 5 groups: groups 1 and 2 were given distilled water orally. Group 3 was administered omeprazole 20 mg/kg, groups 4 and 5 were given ZJP (1 g/kg, 2 g/kg, respectively). After an additional hour, the mice in groups 2–5 received ethanol (0.2 ml/kg) orally while group 1 received distilled water instead. Mice were killed after 4 h and their serum and stomachs subjected to further studies. The superoxide dismutase (SOD) activities and malondialdehyde (MDA) level in serum were assayed by SOD and MDA kits, respectively. The myeloperoxidase (MPO) activities in stomachs were assayed by MPO kit. The levels of inflammatory mediators including tumor necrosis factor-α (TNF-α), interleukin-6 (IL-6) and interleukin-1β (IL-1β) in serum were assayed by enzyme-linked immunosorbent assay method (ELISA). Pathological changes were observed by hematoxylin-eosin (HE) staining. The levels of nuclear factor-κBp65 (NF-κBp65), P-NF-κBp65, P-IκBα, IκBα, P-IKKα, IKKα, P-IKKβ, IKKβ in stomachs were assayed by western blot.

Results: The data showed that treatment with the ZJP markedly attenuated MPO, MDA, TNF-α, IL-6, IL-1β and increased SOD; and ZJP also decreased protein levels of P-NF-κBp65, P-IκBα, P-IKKα and P-IKKβ in gastric stomachs.

Conclusion: It was concluded that ZJP may represent a potential therapeutic option to reduce the risk of gastric ulceration and the gastroprotective activity of ZJP might contribute in adjusting the inflammatory cytokine by regulating the NF-κB signaling pathway.

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1. Introduction

Peptic ulcer is pathological lesions in the gastrointestinal tract that usually occur in the stomach and duodenum (Rocha Caldas et al., 2011). The gastric ulcer is characterized by necrosis, infiltration of neutrophils, reduction in blood flow, induction of oxidative stress, and secretion of inflammatory mediators (Viana et al., 2013). Some of the main aggressive factors that can contribute to ulcer are gastric acid, pepsin, bile salts, abnormal motility, alcohol and nonsteroidal anti-inflammation drugs (NSAID), as well as infection with *Helicobacter pylori*. However, there are several factors that can protect the stomach from ulcer formation such as mucus secretion, gastroprotective prostaglandin

synthesis, bicarbonate production and normal tissue microcirculation (Allen and Garner, 1980; Laine et al., 2008; Peskar and Maricic, 1998; Wallace, 1992). Reducing gastric acid production and increasing gastric mucosal protection have been the major strategies proposed for the prevention of peptic ulcer disease (Hoogerwerf and Pasricha, 2001). Ethanol-induced gastric injury is a commonly known method used in the evaluation of therapeutic potencies against gastric ulcer (Abdelwahab et al., 2013).

It was suggested that, cytokines such as TNF-α, IL-6 and IL-1β play important roles in the acute phase inflammation as well as in maintenance and regulation of the severity of gastric ulcers. Over expression and translocation of the NF-κB subunits (p65 and p50) to

Abbreviations: (ZJP), Zuojin Pill; (SOD), superoxide dismutase; (MDA), malondialdehyde; (MPO), myeloperoxidase; (TNF-α), tumor necrosis factor-α; (IL-6), interleukin-6; (IL-1β), interleukin-1β; (HE), hematoxylin-eosin; (ELISA), enzyme-linked immunosorbent assay method; (NF-κBp65), nuclear factor-κBp65; (TCM), traditional Chinese medicine; (OME), omeprazole; (NSAID), nonsteroidal anti-inflammation drugs

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the nucleus promote the over expression of proinflammatory mediators such as TNF- α and IL-6 (Baldwin, 1996). Increasing evidence reveals that the inhibition of NF- κ B activity may lead to alleviating the severity of inflammatory diseases (Atreya et al., 2008).

Alternative and complementary medicine has gained global attention because of their widespread use in the field of medicine for treating many diseases. Zoujin Pill (ZJP), a typical traditional Chinese medicine (TCM) formula, consists of the Rhizoma Coptidis (*Coptis chinensis* Franch. was used in the study) and Fructus Evodiae (*Tetradium ruticarpum* (A. Juss.) Hartley was used in the study) in the ratio of 6:1 (w/w). ZJP was first recorded in “Danxi’s experiential therapy” for treating gastro-intestinal disorders in the 15th century. It is officially listed in the Chinese Pharmacopoeia as a prescription employed in patients suffering from gastric ulcer, gastroesophageal reflux disease, gastritis, and pyloric obstruction, among other disorders (Pharmacopoeia, 2010). Although ZJP had been found to treat gastro-intestinal disorders, the underlying mechanisms remain unknown. In this study, our objective was to elucidate the effect and the molecular mechanism of ZJP on ethanol-induced gastric ulcer.

2. Materials and methodology

2.1. Reagents

The rhizomes of *C. chinensis* Franch. (Rhizoma Coptidis) and the dried, unripe fruit of *T. ruticarpum* (A. Juss.) Hartley (Fructus Evodiae) were purchased from Xiansheng drug Store (Nanjing, China) and the plant was identified and authenticated by microscopic identification by Prof. Mingjian Qin (China Pharmaceutical University) according to the Chinese Pharmacopoeia (2010). Evodiamine (batch number: 110802–201409) and berberine (batch number: 110713–201212) were purchased from the National Institutes for Food and Drug Control (Beijing, PR China). As the positive control, omeprazole (OME) was supplied by Shanghai Xinyijiahua Pharmaceutical Company Limited (Shannxi, China). SOD and MPO were purchased from the Institute of Jiancheng Bioengineering (Nanjing, China). The enzyme-linked immunosorbent assay (ELISA) kits for determination of IL-6, IL-1 β and TNF- α were produced by Nanjing KeyGEN Biotech. CO., LTD. (Nanjing, China). The anti-P-NF- κ Bp65, NF- κ Bp65, P-I κ B α , I κ B α , P-IKK α , IKK α , P-IKK β , IKK β were supplied by Cell Signaling Technology.

2.2. Ethics statement

All animal experiments were performed according to protocols approved by China Pharmaceutical University (No. CPU-TCM-2013012) Medicine Animal Care and Use Committee.

2.3. Animal

50 male ICR mice (18–22 g) were purchased from the experimental Animal Center of Nanjing Medical University (Nanjing, PR China) (Production license: SCXK2014-0008). The animals were maintained in a temperature controlled room at 20.1–23.11 °C and 40–50% humidity, a 12 h light/dark cycle and free access to water.

2.4. Plant extraction

C. chinensis Franch (120 g) (batch number: 20140901) and *T. ruticarpum* (A. Juss.) Hartley (20 g) (batch number: 20140908) were immersed for 1 h with 8 times volume of distilled water. The medicinal materials were decocted twice at boiling temperature for half an hour, and then the decocted liquids were collected. Decocted liquids were centrifuged at 3000 rpm for 5 min, the

supernatant was concentrated and dried in vacuum at 55 °C. The yield of dried powder was 19.4% according to the original dry materials. The final dried powder of ZJP was dissolved with some distilled water and the final concentration of ZJP was 2 g/ml. The sample was stored at 4 °C. The doses of distilled water extract of ZJP were expressed as gram of the original dry materials per kilogram body weight and the doses (1 g/kg, 2 g/kg) of distilled water extract of ZJP were given to animals according to human dose.

2.5. UPLC-MS analysis of ZJP

2.5.1. UPLC conditions

UPLC was performed on a Waters Acquity UPLC system (Milford, MA), consisting of a binary solvent delivery system, an autosampler and a PDA detector system. An ACQUITY UPLC BEH C18 column (2.1 $\times$ 100 mm² i.d., 1.7 $\times$ 1026 m²; Waters) was used for all analyses. The mobile phase was consisted of A (acetonitrile) and B (aqueous solution of 0.1% phosphoric acid and 0.05% triethylamine). The linear gradient was as follows: 0–7 min, 80% B; 7–12 min, 80–75% B; 12–16 min, 75% B; 16–19 min, 75–45% B, 19–22 min, 45% B at a flow rate of 0.4 ml/min.

2.5.2. Mass spectrometric conditions

The MS analysis was performed on a Waters ACQUITY Synapt Q-TOF mass spectrometer connected to the Waters ACQUITY UPLC system via an electrospray ionization (ESI) interface. High purity nitrogen was used as the nebulizer and auxiliary gas and argon was the collision gas. The Q-TOF MS was operated in negative ion mode with a capillary voltage of 3 kV, a sampling cone voltage of 30 V, a cone gas flow of 50 L/h, a desolvation gas flow of 600 L/h, a desolvation temperature of 3508 °C, a source temperature of 1208 °C, a collision energy of 6 V and full scan spectra from 100 to 1000 Da.

2.6. Induction of gastric ulcer and treatment

Five groups of mice were assigned (Li et al., 2014): groups 1 (control) and 2 (model) were given distilled water orally. Group 3 was administered omeprazole (OME) 20 mg/kg, groups 4 and 5 were given ZJP (1 g/kg, 2 g/kg, respectively) (Chen et al., 2009; Li et al., 1993). All drugs were administered once daily for 4 days. Drugs were given by gastric gavage, once daily and were suspended in saline. On the last day of treatment, after an additional hour, the mice in groups 2–5 received ethanol (0.2 ml/kg) orally while group 1 received distilled water instead. Mice were euthanized with 10% chloral hydrate after 4 h and their serum and stomachs subjected to further studies.

2.7. Measurement of gastric juice acid content (pH)

The gastric contents of each mouse were collected and centrifuged at 4000 rpm for 10 min. The pH of the supernatant for each sample was measured using a pH meter.

2.8. MPO, MDA and SOD levels

MPO activity in stomachs, MDA and SOD levels in serum were determined using test kits purchased from Nanjing Jiancheng Bioengineering Institute (China, Nanjing); all procedures were according to the manual.

2.9. Cytokine measurements

Serum levels of IL-6, TNF- α and IL-1 β were measured by ELISA according to the manufacturer’s instructions (R&D, Minneapolis, MN, USA). All measurements were performed in duplicate.

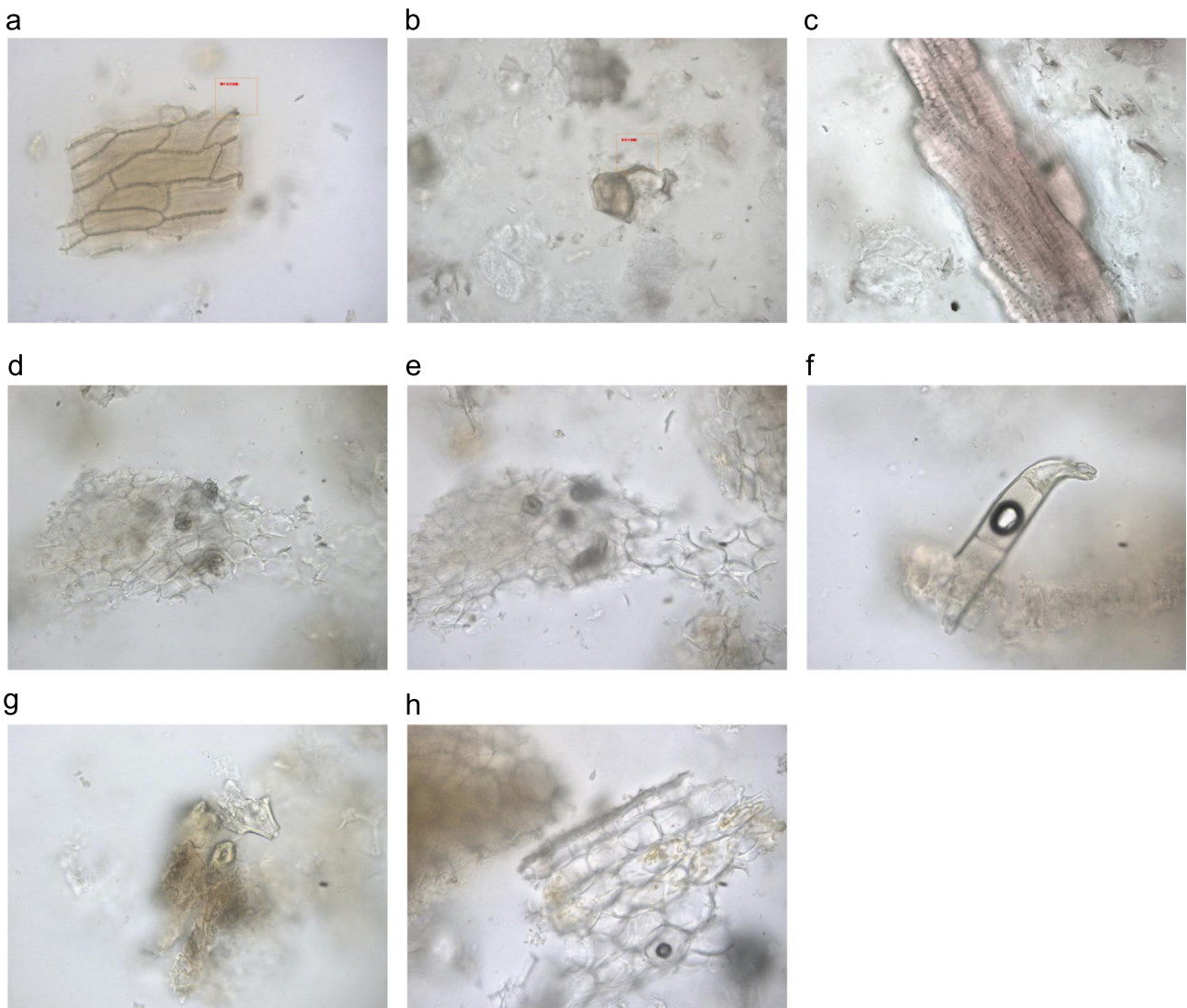


Fig. 1. Microscopic identification of *Coptis chinensis* Franch. (Rhizoma Coptidis) and *Tetradium ruticarpum* (A. Juss.) Hartley (Fructus Evodiae). (1) The stone cell (a); (2) the epidermal cells like scales (b); (3) pericycle fibers (c) in *C. chinensis* Franch. (Rhizoma Coptidis), (2) Calcium oxalate crystal clusters (d); (2) calcium oxalate square crystal cluster (e); (3) non glandular hairs (f); (4) the stone cell (g); (5) the oil chamber pieces (h) in *T. ruticarpum* (A. Juss.) Hartley (Fructus Evodiae)

2.10. Histological procedure and assessment

For histological assessment, the glandular stomach was fixed in 10% neutral buffered formalin solution for 24 h, sectioned, and then embedded in paraffin. 5 mm sections thick were deparaffinized, stained with hematoxylin–eosin (H&E), and examined under a light microscope. The specimens were then assessed according to the criteria of [Laine and Weinstein \(1988\)](#). Briefly, a 1 cm segment of each histological section was assessed for epithelial cell loss (score: 0–3), edema in the upper mucosa (score: 0–4), hemorrhagic damage (score: 0–4), and the presence of inflammatory cells (score: 0–3), yielding a maximum total score of 14. Afterward, the sections were assessed by an experienced pathologist who was blinded to the treatment.

2.11. Western blot

Stomach tissue was chopped into small pieces and homogenized in ice-cold RIPA buffer containing 0.1% phenylmethylsulfonyl fluoride. The dissolved proteins were collected from the supernatant after centrifugation at 12,000 g for 20 min. Protein

concentrations were determined using Coomassie blue based assay reagent. Protein extracts were separated by a SDS-polyacrylamide gel electrophoresis and then transferred onto a PVDF membrane. The membrane was blocked with 5% skim milk in Tris buffer saline and then incubated at 4 °C overnight with respective primary antibodies for anti-NF- κ Bp65, P-NF- κ Bp65, P-I κ B α , I κ B α , P-IKK α , IKK α , P-IKK β and IKK β . After washing with tris-buffered saline–tween 20 (TBST), the membranes were incubated with a horseradish peroxidase conjugated secondary antibody (1:12,000) for 1.5 h at room temperature. The antibody-reactive bands were visualized by using enhanced chemiluminescence detection reagents and a gel imaging system (Tanon Science & Technology Co., Ltd., China).

2.12. Statistical analysis

All data were normally distributed and are presented as mean $\pm$ S.E.M. In the case of single mean comparison, data were analyzed by a Student's test. In the case of multiple mean comparisons, the data were analyzed by ANOVA and the Newman–Keuls post test, or two-way repeated measures ANOVA, followed by Bonferroni multiple

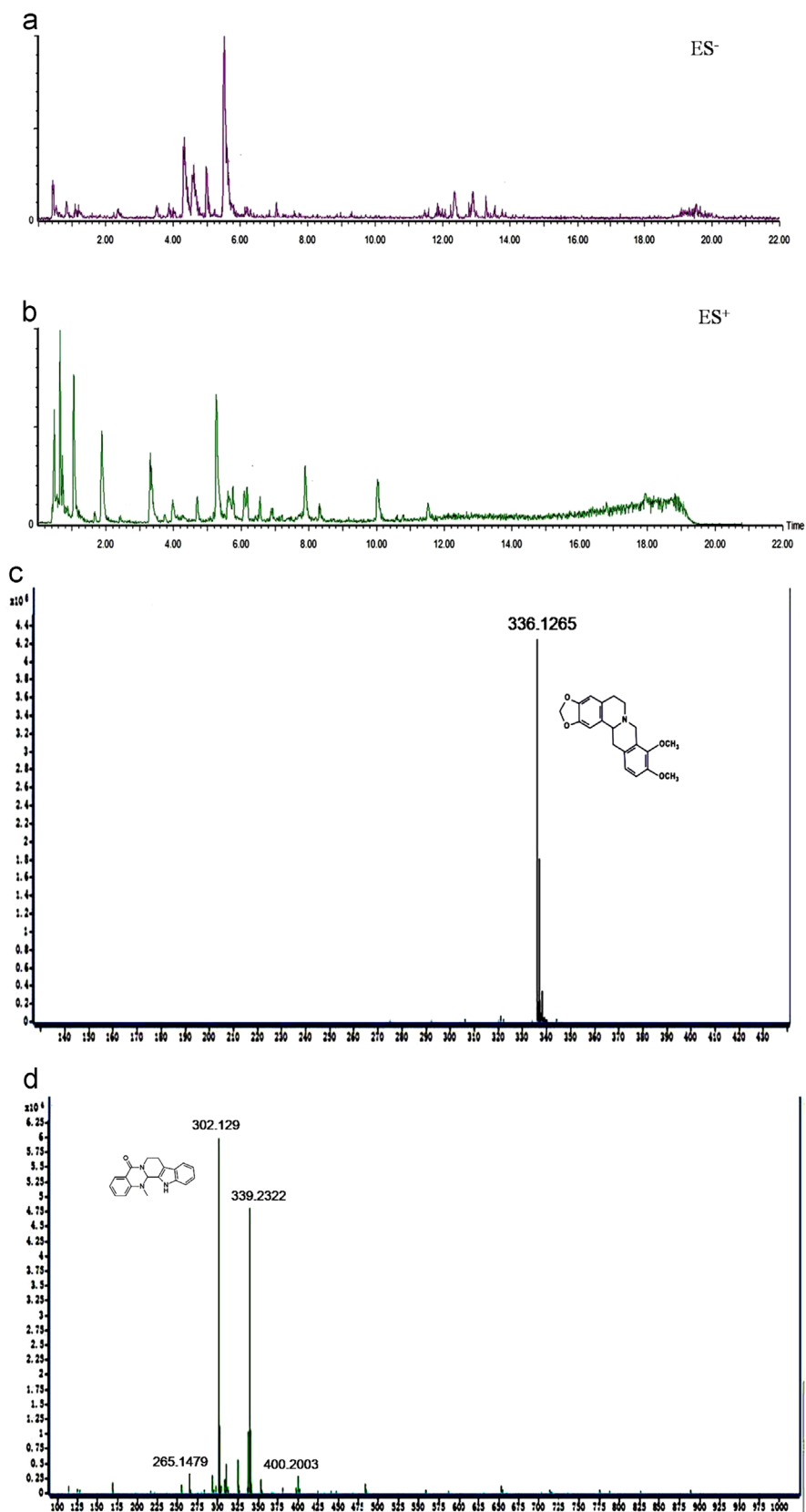


Fig. 2. UPLC-MS profile of ZJP. (a) Representative chromatograms of ZJP in negative ion mode, (b) representative chromatograms of ZJP in positive ion mode, (c) representative mass spectra of berberine, and (d) representative mass spectra of evodiamine.

comparison tests. *P*-values less than 0.05 were regarded to reflect a significant difference.

3. Results

3.1. Microscopic identification of *Coptis chinensis* Franch. (Rhizoma *Coptidis*) and *Tetradium ruticarpum* (A. Juss.) Hartley (Fructus *Evodiae*)

The *C. chinensis* Franch. (Rhizoma *Coptidis*) and the *T. ruticarpum* (A. Juss.) Hartley (Fructus *Evodiae*) were identified and authenticated by Prof. Mingjian Qin (China Pharmaceutical University) according to the Chinese Pharmacopoeia (2010). The following are characteristics of the *C. chinensis* Franch. (Rhizoma *Coptidis*) and the *T. ruticarpum* (A. Juss.) Hartley (Fructus *Evodiae*) (Fig. 1):

The characteristics of *C. chinensis* Franch. (Rhizoma *Coptidis*) are follows: (1) the stone cell (a); (2) the epidermal cells like scales (b); (3) pericycle fibers (c).

The characteristics of *T. ruticarpum* (A. Juss.) Hartley (Fructus *Evodiae*) are as follows: (1) calcium oxalate crystal clusters (d);

(2) calcium oxalate square crystal cluster (e); (3) non glandular hairs (f); (4) the stone cell (g); (5) the oil chamber pieces (h).

3.2. UPLC-MS profile of ZJP

The major components of ZJP were analyzed by UPLC-MS (Fig. 2). Two compounds were identified and determined: berberine and evodiamine, which is consistent with previous study (Gao et al., 2010). The contents of these compounds in ZJP were observed as: 6.33 mg/g, 1.26 mg/g.

3.3. Effect of ZJP on gastric juice acid content (pH)

Significant decreases in PH observed in the model mice, compared with the control mice. Pretreatment with ZJP increased PH compared with model mice a dose-dependent manner (Fig. 3).

3.4. Effect of ZJP on MPO, MDA and SOD levels

The levels of MPO, MDA were increased and the levels of SOD decreased in model (Fig. 4). ZJP (1 g/kg, 2 g/kg) can down-regulate the MDA, MPO levels and up-regulate the SOD.

3.5. Effects of ZJP on cytokine in serum

The levels of the cytokines TNF- α , IL-6 and IL-1 β in serum were dramatically increased in model (Fig. 5). ZJP (1 g/kg, 2 g/kg) can down-regulate the TNF- α , IL-6 and IL-1 β level in dose-dependent manners.

3.6. Histology of gastric lesions

Gastric ulcer examination by H&E staining showed clear gastric mucosal damage. Some necrotic lesions also penetrated deeply into the mucosa of the ulcerated stomachs of the ulcer group. Other effects due to ethanol pre-treatment were leukocyte infiltration and extensive edema of the submucosal layer. However,

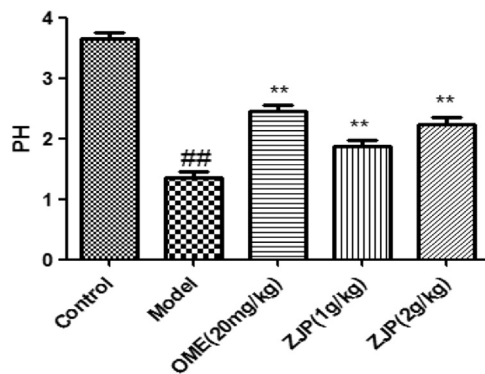


Fig. 3. Effect of ZJP on gastric juice acid content (pH). All values given are the mean $\pm$ SEM. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ vs. control group. $^{*}P < 0.05$ and $^{**}P < 0.01$ vs. model group.

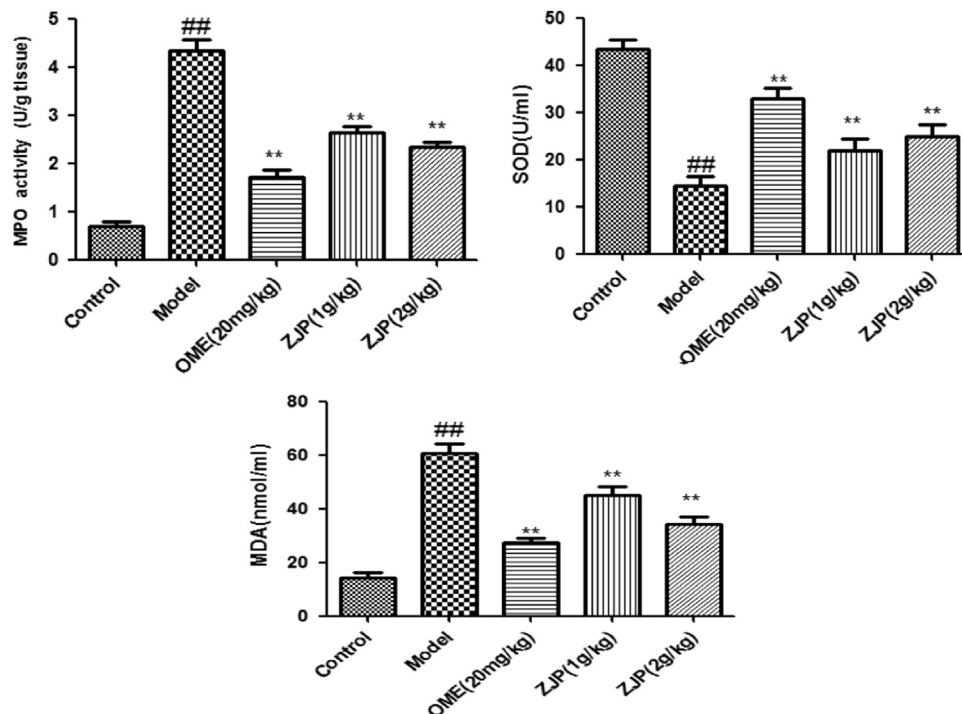


Fig. 4. Effect of ZJP on MPO, MDA and SOD levels. All values given are the mean $\pm$ SEM. $^{\#}P < 0.05$ and $^{\#\#}P < 0.01$ vs. control group. $^{*}P < 0.05$ and $^{**}P < 0.01$ vs. model group.

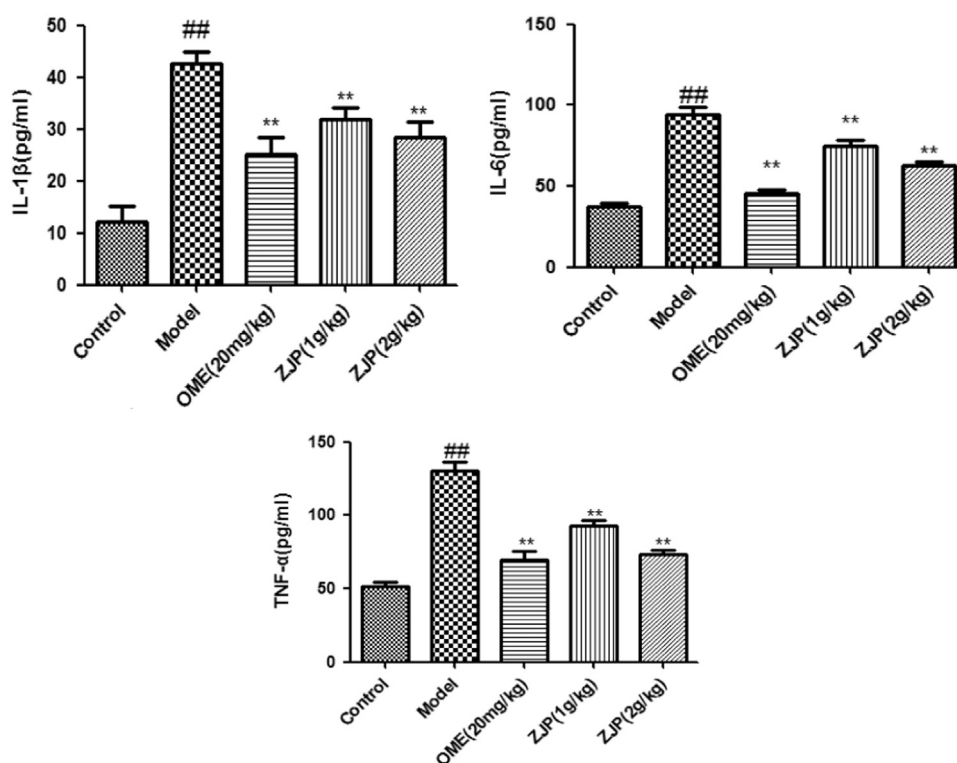


Fig. 5. Effects of ZJP on cytokine in serum. All values given are the mean $\pm$ SEM. $^{\#}P < 0.05$ and $^{##}P < 0.01$ vs. control group. $^{*}P < 0.05$ and $^{**}P < 0.01$ vs. model group.

groups pre-treated with ZJP exhibited remarkable protection via reduction of the ulcer area, leukocyte infiltration and submucosal edema (Fig. 6).

3.7. Effects of ZJP on NF- κ Bp65, P-NF- κ Bp65, P-I κ B α , I κ B α , P-IKK α , IKK α , P-IKK β and IKK β

The levels of the P-NF- κ B, P-I κ B α , P-IKK α and P-IKK β in stomachs were dramatically increased in model (Fig. 7), ZJP (1 g/kg, 2 g/kg) can down-regulate P-NF- κ B, P-I κ B α , P-IKK α and P-IKK β .

4. Discussion

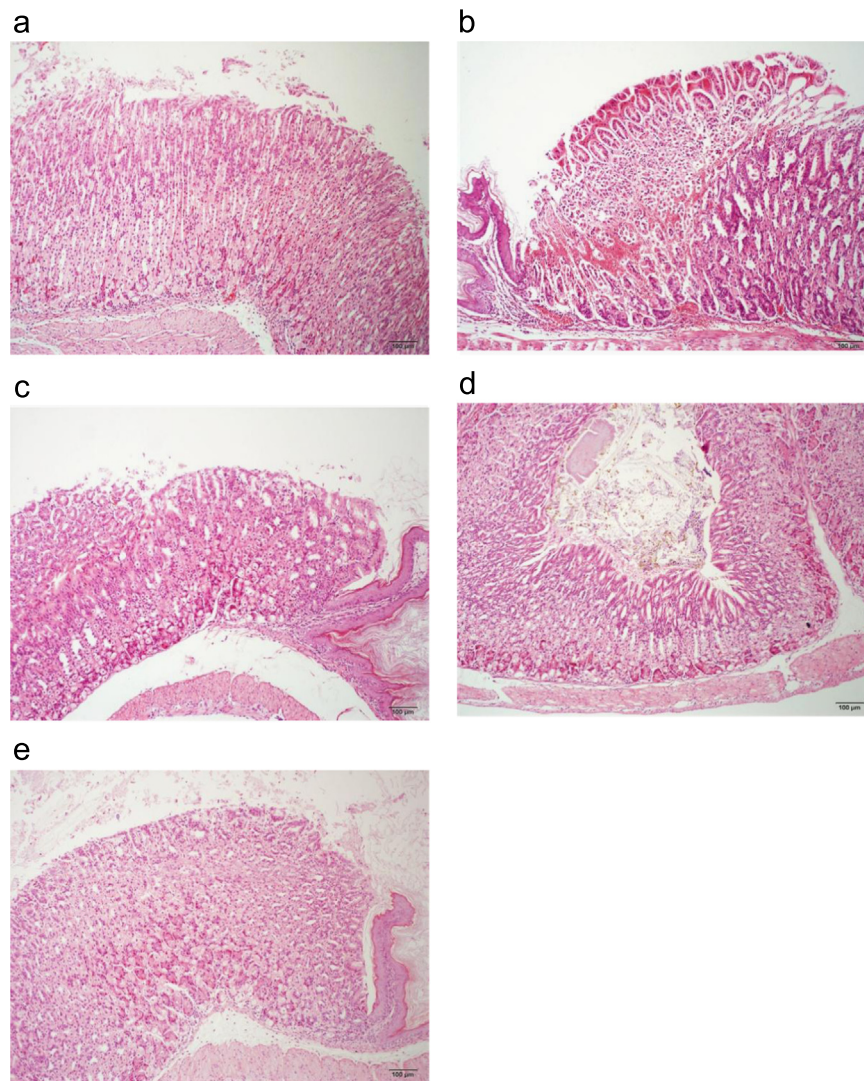
We investigated the gastroprotective activity of ZJP in a model of ethanol-induced experimental gastric injury. Ethanol consumption can produce acute inflammatory reaction. However, oral administration of ZJP effectively reverses ethanol-induced gastric injury. The protective activity of ZJP was ascertained by comparing the gastric damage caused by ethanol to the mice in the ulcer control group with the stomachs of the mice in the normal and the treatment groups.

The potential antiulcer drugs can exert their protective effect against mucosal lesions through inhibition of neutrophil infiltration in the ulcerated gastric tissue (Ganguly et al., 2005). Furthermore, neutrophil accumulation in a gastric mucosa has been shown to induce microcirculatory abnormalities. The neutrophil infiltration into the gastric mucosal tissues is assessed by MPO activity, because MPO assay has been wide spread used as an index of neutrophil infiltration in various experimental gastric injuries, colitis, and human gastric ulcer (Nishida et al., 1998). In the present study, we observed a significant increase in MPO activity in the stomach following ethanol administration, which confirmed the infiltration and activation of neutrophils in the gastric mucosa produced by ethanol. ZJP pre-treatment in the animals with gastric

ulcer significantly reduced the levels of MPO, suggesting the ability of these preparations in preventing the neutrophil infiltration in ulcerated gastric tissue.

The possible changes in the level of cytokines IL-6 and TNF- α were investigated. The observed increased in the production of IL-6 and TNF- α in serum and ulcerated gastric tissue can be afforded to the necrotizing effects of ethanol (Sivalingam et al., 2008). Proinflammatory cytokines, particularly TNF- α , are well-known inducers of apoptotic cell death by activating the caspase family of proteases (Tsukimi and Okabe, 2001). Besides many biological activities, TNF- α is involved in the induction of inflammation, injury and carcinogenesis in a variety of tissues including the gastric mucosa (Faubion and Gores, 1999). Moreover, TNF- α has overlapping and synergetic activities to induce the production of nuclear factor kappa B (NF- κ B) and other cytokines (Zhou et al., 2006). Elevated levels of IL-6 activates neutrophils, lymphocytes and monocytes/macrophages at the inflammatory site, which in turn initiates different oxidative bursts, toxic metabolites and lysosomal enzymes responsible for local tissue damage in peptic ulcer disease. TNF- α and IL-6 levels in serum and gastric mucosa correlate with the level of gastric mucosal inflammation. However, the results of this study demonstrated the ability of ZJP to inhibit the level of proinflammatory cytokines mediator (TNF- α and IL-6), which showed its anti-inflammatory effect in ethanol-induced gastric ulcer.

It has been demonstrated that these pro-inflammatory mediator are regulated by NF- κ B pathway. Therefore, NF- κ B is considered an ideal target for the molecular therapy of gastric ulcer. Protein kinase C (PKC) has been shown to play a dual role in activation of NF- κ B in a number of model systems. Notably, PKCs was able to phosphorylate and activate I κ B (Diaz-Meco et al., 1994), thus demonstrating its direct implication in the NF- κ B activation process by participating in I κ B degradation. Another mechanism by which PKCs regulates NF- κ B may include phosphorylation of the p65 subunit (Rahman et al., 2000; Anrather et al., 1999). Therefore, we hypothesized that ZJP, could exert its gastroprotective and ulcer healing



Groups	Dose(g/kg)	Hemorrhagic damage (score 0–4)	Edema (score 0–4)	Epithelial cell loss (score 0–3)	Inflammatory cells (score 0–3)
Control	-	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00
Model	-	3.36±0.27##	3.43±0.42##	3.27±0.13##	2.57±0.18##
OME	2×10 ⁻²	1.21±0.22**	1.46±0.36**	1.36±0.18**	1.26±0.19**
ZJP	1	2.41±0.25**	2.37±0.26**	2.58±0.18**	1.76±0.17**
	2	2.14±0.42**	1.78±0.31**	2.17±0.15**	1.57±0.13**

Fig. 6. Histology of gastric lesions. All values given are the mean ± SEM. #*P* < 0.05 and ##*P* < 0.01 vs. control group. **P* < 0.05 and ***P* < 0.01 vs. model group. (A) Control; (B) model; (C) OME; (D) ZJP (1 g/kg); (E) ZJP (2 g/kg).

actions effect on gastric ulcers by inhibiting the activation of NF-κB signaling pathway. In the present study, we firstly discovered that anti-inflammatory effect of ZJP may be linked with down regulation of NF-κB p65 activity in ethanol-induced gastric ulcer in mice. Western blot test result indicated that protein expression of NF-κB p65 significantly increased in ethanol-induced gastric ulcer in mice and obviously decreased in groups after continuous 4-day

treatment with ZJP. On the other hand, our experiment indicated that the levels of TNF-α and IL-6 in gastric ulcer group increased more distinctly than normal group and reduced dose-dependently after treatment with ZJP in response to NF-κBp65. Since the promoter regions of TNF-α and IL-6 had also been shown to contain consensus binding motifs for NF-κB. We presume that ZJP may inhibit through suppression of NF-κB activation.

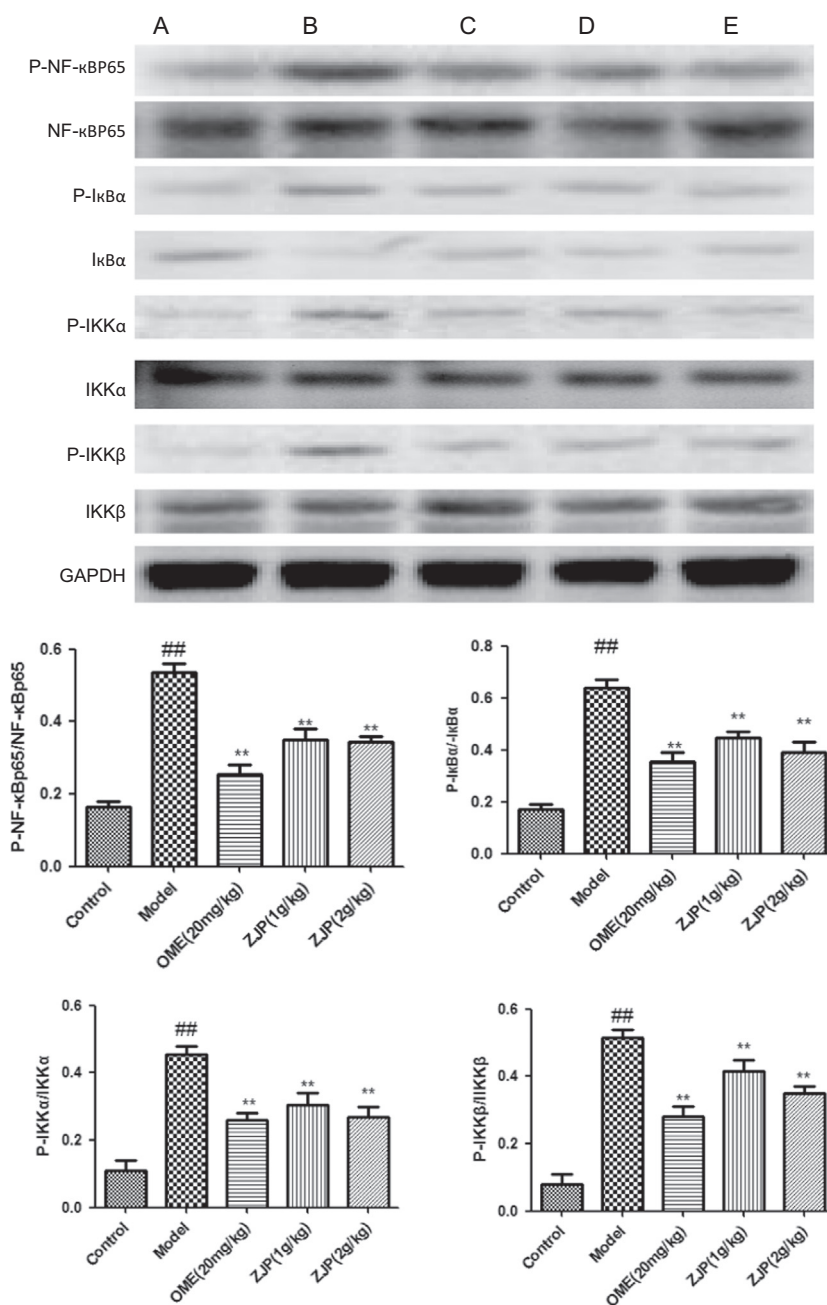


Fig. 7. Effects of ZJP on NFκB pathway. All values given are the mean ± SEM. [#]*P* < 0.05 and ^{##}*P* < 0.01 vs. control group. ^{*}*P* < 0.05 and ^{**}*P* < 0.01 vs. model group. (A) Control; (B) model; (C) OME; (D) ZJP (1 g/kg); (E) ZJP (2 g/kg).

In conclusion, the results of the present report demonstrate that ZJP pre-treatment exert beneficial effects in mice model of gastric ulcer induced by the ethanol. We observed that ZJP administration significantly decreased MPO activity, and reduced expression of pro-inflammatory mediators IL-6 and TNF-α in serum in mice exposed to ethanol, these protective effects were associated with a reduced level of NF-κB pathway. This study might be limit and further and comprehensive studies are needed in the future.

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